WEEK 1 · LESSON 5 OF 24 · HANDOUT

Search and Problem Solving

Study time: **about 50 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=ySN5Wnu88nE — open the lesson online to watch inside the course.

🎓 **Go deeper:** Harvard CS50 AI, Lecture 0: Search — https://www.youtube.com/watch?v=WbzNRTTrX0g

Your goal in this lesson: Describe a problem as a search problem, explain how breadth-first, depth-first and A* search work, and say what a heuristic is.

Many problems are search problems

Finding a route on a map, solving a puzzle, scheduling exams or planning a delivery all share one shape: there is a **start**, a **goal**, and many possible **steps** in between. The computer must search for a good path. This is one of the oldest ideas in AI, and it still powers maps apps and logistics today.

The parts of a search problem

- **State** — a description of the situation (*the bus is in Muhanga*).
- **Initial state** — where we start (*Kigali*).
- **Actions** — what we can do from a state (*drive to a neighbouring town*).
- **Goal test** — how we know we are done (*are we in Rubavu?*).
- **Path cost** — what a path costs (distance, time, fuel).

All the states and the actions between them form a **state space**, often drawn as a graph: towns are dots, roads are lines.

Uninformed search: no hints

Breadth-first search (BFS) explores all places one step away, then all places two steps away, and so on — like ripples on water. - Finds the path with the **fewest steps**. - Uses a lot of memory on big problems.

Depth-first search (DFS) follows one road as far as it can, then backs up and tries another. - Uses little memory. - May find a long, poor path first, or get lost in very deep spaces.

Uniform-cost search always expands the cheapest path so far. It finds the **lowest-cost** path even when steps have different costs (some roads are longer).

Informed search: using a hint

A **heuristic** is an estimate of how far a state is from the goal. On a map, the straight-line ("as the crow flies") distance to the destination is a good heuristic: it is easy to compute and never more than the real road distance.

A-star search (written `A*`) ranks each option by:

> ***f* = *g* + *h*** *— cost so far (g) plus estimated cost still to go (h)*

It expands the option with the smallest *f*. If the heuristic never **overestimates** the real remaining cost, A-star is guaranteed to find the cheapest path, and it usually explores far fewer states than uninformed search. Route-planning software uses ideas from A-star every day.

Worked example

From Kigali you can go to Muhanga (50 km) or Musanze (90 km). You want to reach Rubavu. Straight-line estimates to Rubavu: Muhanga 100 km, Musanze 55 km.

- Muhanga: f = 50 + 100 = 150
- Musanze: f = 90 + 55 = 145

A-star explores Musanze first, because it looks cheaper overall — even though the first step is longer. That is the power of combining real cost with a good estimate.

Search in games

In two-player games (chess, draughts), the agent must expect the opponent to fight back. **Minimax** search assumes the opponent always picks the move worst for you, and chooses the move with the best guaranteed result. Deep Blue (1997) used this kind of search with very fast hardware.

Key terms

- **State space** — all possible states and the actions linking them.
- **Heuristic** — a quick estimate of the distance to the goal.
- **Admissible heuristic** — one that never overestimates; makes A-star optimal.
- **Optimal** — finds the best (cheapest) solution.

Remember this: Search turns a problem into states and actions. Uninformed search explores blindly; informed search like A-star uses a heuristic (f = g + h) to reach good answers faster.

Name: _____ Date: _____

Activity

Task 1

Draw a map with six places you know (home, school, market, church, bus park, hospital) and the walking minutes between them.

Task 2

Run breadth-first search by hand from home to the hospital and write the order you visit places.

Task 3

Invent a heuristic for each place (your best guess of minutes to the hospital). Run A-star and compare the path and the number of places explored.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which is NOT a part of a search problem?
   A. Initial state
   B. Path cost
   C. Goal test
   D. Screen brightness

2. Breadth-first search guarantees the path with…

A. Random steps
B. The most steps
C. The highest cost
D. The fewest steps

3. What is a heuristic?

A. A list of all states
B. A quick estimate of how far a state is from the goal
C. The exact final answer
D. A type of computer

4. In A* search, what does f = g + h mean?

A. Cost so far plus estimated cost to go
B. Final score = games + hours
C. Height plus width
D. Goal minus heuristic

5. When is A* guaranteed to find the cheapest path?

A. Never
B. When the heuristic never overestimates the real remaining cost
C. When it uses depth-first order
D. When the heuristic is always zero only

6. Kigali → Muhanga costs 50 with h=100; Kigali → Musanze costs 90 with h=55. Which does A* expand first?

A. Neither
B. Musanze, because 145 < 150
C. Muhanga, because 50 < 90
D. Both at once

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